TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INCORPORATED (TELTECH)

NATIONWIDE 5G STANDALONE (SA) FIXED WIRELESS ACCESS (FWA) NETWORK

RFP REFERENCE: RFP-TELTECH-FWA-2026-001

BID BULLETIN NO. 2.1: COMPREHENSIVE TECHNICAL, OPERATIONAL & COMMERCIAL ADDENDUM

TO ALL PROSPECTIVE BIDDERS: All Qualified Bidders, System Integrators, and Authorized Equipment Manufacturers

FROM: Oliver Tungol, Lead Technical Architect & Tender Secretariat

DATE OF ISSUANCE: October 30, 2026 (Revised & Consolidated Addendum)

SUBJECT: Technical Baseline Revisions, Mandatory RF Conformance Directives (RAN-21), and Promulgation of Consolidated 100-Question Clarifications Master

> **GOVERNING LEGAL DIRECTIVE:** This **Bid Bulletin No. 2.1** is formally promulgated pursuant to Section 1.5 (*Clarification and Amendment of Bidding Documents*) of the Master RFP Document. It updates, refines, and consolidates the directives of Bulletin 2.0. The provisions, technical standards, mandatory RF conformance testing protocols (RAN-21), and commercial rules articulated herein are seamlessly codified as binding contractual addenda to Documents 01 through 09. Prospective bidders must formulate their final 3-envelope submissions strictly in conformance with these unified engineering policies.

1. EXECUTIVE SUMMARY & TENDER STATUS OVERVIEW

Following detailed technical reviews, preliminary engineering evaluations, and structural harmonization across the 5,000 candidate barangays, TelTech officially issues this comprehensive policy bulletin (Revision 2.1). This

document codifies authoritative directives across nine (9) strategic technical domains:

1. **Cloud-Native 5G SA Core Dimensioning & User Plane Throughput (Lot 3)**
2. **Spectrum Allocation (NTC MC 006-11-2025), Band n50 Operating Passband, RF Conformance Testing (RAN-21) & Demarcation Optics (Lot 1 & Lot 6)**
3. **Customer Premises Equipment (CPE) Phased Rollout & Lifecycle Reverse Logistics (Lot 7)**
4. **Civil Works, Monopole Structural Standards & Site Power Infrastructure (Lots 2 & 5)**
5. **Operational SLAs, Radio Network KPIs & SSV Acceptance Standards (Document 04 / Appendix C)**
6. **Digital OSS/BSS, Field Force Automation & VAS Subsystems (Lot 4)**
7. **Candidate Siting Schedule Precision & Geographic Asset Provenance (Document 07)**
8. **Commercial BoQ Harmonization, Multi-Year O&M & 10-Year TCO Rules (Document 06)**
9. **Consolidated Response Matrix (Part 3: Inquiries Q045 – Q100 & Publication of 100-Question Master Register in Document 09)**

Digital OSS/BSS, Field Force Automation & VAS Subsystems (Lot 4)

Candidate Siting Schedule Precision & Geographic Asset Provenance (Document 07)

Commercial BoQ Harmonization, Multi-Year O&M & 10-Year TCO Rules (Document 06)

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2. CLOUD-NATIVE 5G CORE (5GC SA) & DATA CENTER ARCHITECTURE (LOT 3)

2.1 User Plane Throughput Capacity Dimensioning

1. **Phase 1 Capacity Baseline (1,000,000 Subscribers):**
   - The cloud-native 5G SA Core User Plane Function (UPF) cluster must be physically and logically dimensioned to deliver an initial aggregated user-plane throughput floor of **300 Gbit/s to 500 Gbit/s**.
   - Dimensioning derives from an initial target of 300,000 concurrently active broadband households exhibiting an Average Busy-Hour Throughput (ABHT) of 1.5 Mbit/s to 2.0 Mbit/s per active subscriber at a standard 15% to 20% busy-hour concurrency factor.
2. **Expansion Horizon (3,000,000 Subscribers):**
   - The proposed containerized 5G Core Network Functions (CNF) running on Kubernetes / bare-metal server infrastructure must natively support horizontal auto-scaling of distributed UPF instances up to an aggregated user-plane throughput capacity of **1.5 Tbit/s**.
   - Expansion to 1.5 Tbit/s must be executable via software license expansion and incremental server node deployment without requiring architectural redesign, control plane migration (AMF/SMF/UDM), or database re-sharding.

3. RADIO ACCESS NETWORK (RAN), BAND n50 SPECTRUM & TRANSMISSION (LOTS 1 & 6)

3.1 Spectrum Physics, Operational Channel Window, RF Testing & Emission Suppression

1. **Active Operational Passband & Numerology:**
   - Under the National Telecommunications Commission (NTC) regulatory framework for dedicated broadband wireless communications, the authorized operational Band n50 carrier is established as **1457.0 MHz to 1517.0 MHz**, providing a single contiguous **60.0 MHz TDD channel bandwidth** centered at **1487.0 MHz** (aligned on the standard 100 kHz 3GPP RF reference raster).
   - The carrier operates at 30 kHz Subcarrier Spacing (SCS, μ=1), dimensioned for **162 Physical Resource Blocks (PRBs)** occupying 58.32 MHz active transmission bandwidth.
   - **Spectral Efficiency Comparison:** A single 60.0 MHz carrier provides 162 PRBs (97.2% frequency utilization), capturing +5.9% higher usable capacity and eliminating inter-carrier guard band waste and Carrier Aggregation (CA) overhead compared to 3x 20 MHz (153 PRBs total, 91.8% utilization).
2. **Guardband Allocation & Satellite Protection (> 1517.0 MHz):**
   - A dedicated **1.0 MHz external guardband (1517.0 MHz to 1518.0 MHz)** is allocated immediately adjacent to the upper band edge below the NTC allocation ceiling (1518.0 MHz).
   - Incorporating the internal 3GPP channel guardband of 840 kHz, the effective physical separation from the outermost active subcarrier (1516.16 MHz) to the sensitive satellite downlink boundary (1518.0 MHz) is **1.84 MHz**.
   - All gNodeB transceivers must implement sharp digital filtering ensuring out-of-band emissions (OOBE) into the adjacent maritime mobile-satellite service band (1518.0 – 1525.0 MHz) do not exceed **-30 dBm/MHz** at the antenna connector, strictly compliant with **ITU-R M.2159-0 / WRC-15 Resolution 223** and **3GPP TS 38.104 Category B**.
3. **Lower Out-of-Band Emission Limits (1432.0–1457.0 MHz):**
   - Radiated and conducted unwanted emissions in the adjacent 1432.0–1457.0 MHz block (allocated for Non-Public Networks / NPN under NTC MC 006-11-2025) must strictly meet **3GPP TS 38.104 Category B** standards with an Adjacent Channel Leakage Ratio (**ACLR ≥ 45 dB**).
4. **In-Band Broadcasting-Satellite Service (BSS) Coexistence (1452.0–1492.0 MHz):**
   - In accordance with Section IV of NTC MC No. 006-11-2025, gNodeB macro deployments overlapping the 1452.0–1492.0 MHz primary BSS allocation must support remote power control, electrical downtilt adjustments (eRET), and interference coordination to avoid detrimental interference to licensed satellite earth stations.
5. **Mandatory 3GPP TS 38.104 & TS 38.141 RF Conformance Testing Deliverables:**

- Shortlisted RAN bidders are formally required to submit certified third-party or OEM factory laboratory test reports under **3GPP TS 38.141-1 (Conducted)** and **TS 38.141-2 (Radiated)** demonstrating:
  - **Spectrum Emission Mask (SEM):** Full compliance operating at maximum rated transmitter power (+55 dBm / 320 W conducted per 8T8R sector).
  - **Adjacent Channel Leakage Ratio (ACLR):** Measured at full load (162 PRBs active).
  - **Operating Band Unwanted Emissions (OBUE):** High-resolution spectral emission mask plots across the critical **1517.0 – 1559.0 MHz** range (specifically proving the ≤ -30 dBm/MHz limit above 1518.0 MHz).
  - **Spurious Emissions:** Measured conducted unwanted emissions outside the operating band.
  - **Power De-Rating & Filtering Declaration:** If power back-off or an optional external high-Q cavity bandpass filter is required to satisfy emissions at the 1518.0 MHz boundary, the bidder must explicitly declare the exact de-rating (in dB) and quantify its impact on cell-edge link budget coverage.

6. **NTC Memorandum Circular No. 006-11-2025 & Spectrum Expansion Clarification:**
   - **Regulatory Allocation Framework:** The National Telecommunications Commission (NTC) has promulgated NTC Memorandum Circular No. 006-11-2025 governing International Mobile Telecommunications (IMT) spectrum in the 1427.0–1518.0 MHz (1.5 GHz L-Band) frequency range:
     - **1427.0 – 1457.0 MHz (30 MHz):** Allocated for **Non-Public Networks (NPN)** / dedicated private wireless networks.
     - **1457.0 – 1518.0 MHz (61 MHz):** Allocated for **Public Networks (PN)**, formally confirming the regulatory foundation for TelTech's authorized 1457.0–1517.0 MHz (60 MHz continuous TDD) active operational carrier.
   - **Regulatory Consultations on Spectrum Expansion:** The Operator is actively conducting regulatory and technical consultations with the NTC to clarify mechanisms for expanding operational spectrum bandwidth beyond the 60 MHz baseline (up to 80 MHz channel operation). Specifically, TelTech is clarifying whether supplemental operational access or secondary co-channel/adjacent-channel usage may be authorized, particularly across suburban, provincial, and rural clusters where:
     1. No Non-Public Networks (NPN) are licensed or physically operating within the 1427.0–1457.0 MHz block; and/or
     2. Incumbent mobile network operators (Globe Telecom, Smart Communications) are not actively utilizing the L-Band / Band n75 as Supplemental Downlink (SDL).
   - **Technical Bidding Implications (Hardware & Software):**
     - **Hardware Wideband Native Capability:** All proposed 5G NR Baseband Units (BBU), Remote Radio Units (RRU) / Active Antenna Units (AAU), and Customer Premises Equipment (CPE) RF front-ends must natively support the full 3GPP Band n50 tuning range (**1432.0 – 1517.0 MHz, 85 MHz Instantaneous Bandwidth / IBW**) in hardware.
     - **Software Reconfigurability:** Should the NTC grant supplemental operational bandwidth or regional secondary channels, systems must support software/firmware reconfiguration (e.g., expanding from 60 MHz to 80 MHz channel bandwidth or activating carrier aggregation) remotely without requiring physical antenna swaps, hardware replacements, or tower truck rolls.

3.2 Baseband Architecture & Form Factor Preferences

1. **Form Factor Preference for Co-Location (897 Sites):**
   - Outdoor blade Baseband Units (blade BBUs) featuring zero ground-footprint pole/wall mounting are strongly preferred across all 897 tower co-location sites. Blade architectures minimize TowerCo compound lease fees and accelerate deployment velocity. Compact outdoor cabinet BBUs are permitted only where indoor shelter space is pre-contracted.
2. **Site Transport Capacity & Optical Transceiver Boundary:**
   - Initial gNodeB backhaul throughput dimensioning of **1 Gbit/s** per site is approved for Day 1 traffic modeling.
   - However, backhaul interfaces must support hardware upgradeability to **10 Gbit/s** via standard SFP+ optical cages.
   - **Demarcation Supply Scope:** The RAN contractor shall supply dual-rate **10GE/25GE eCPRI/Ethernet adaptive optical modules** (1310nm Single-Mode Fiber, LC duplex, 10 km reach) for all BBU optical ports. TelTech shall provide compatible optical transceivers on the transmission demarcation router/switch ports.
   - **Configuration Boundaries:** The RAN contractor executes all BBU port IP address provisioning, VLAN tagging, QoS DSCP priority mapping, and IEEE 1588v2 PTP / SyncE frequency/phase synchronization. TelTech transmission teams configure transport tunneling (MPLS-VPN / SRv6).

3.3 Nearest Tower Geographic Coordinates & Siting Inventory Precision (Document 07)

1. **Candidate Siting Geographic Precision:**
   - In response to technical simulation requirements, high-precision WGS84 geographic coordinates (6 decimal places) for all nearest co-location towers have been populated in Document 07 (`07_RFP_5G_FWA_Candidate_Site_List_and_Rollout_Schedule.xlsx`) across Tab 'All 5,000 Sites' and Batch 1 through Batch 5.
   - All 60 previously flagged reversed coordinate entries have been fully verified and corrected against audited telecom asset databases.
   - Bidders are reminded that tower coordinate data constitute confidential operator assets provided under the RFP Non-Disclosure Agreement (NDA) solely for RF propagation modeling, link budget simulations, and construction feasibility assessment.

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4. CUSTOMER PREMISES EQUIPMENT (CPE) ROLLOUT & LIFECYCLE REVERSE LOGISTICS (LOT 7)

4.1 Phased Hardware Delivery Model

1. **Initial Phase 1 Form Factor:**
    - Recognizing commercial manufacturing timelines for Band n50 outdoor units, TelTech authorizes a phased rollout: Bidders are permitted to supply high-gain 5G Indoor Units (IDU, minimum 8.0–10.0 dBi equivalent antenna gain with integrated 4x4 MIMO and external TS-9/SMA antenna ports) for initial Phase 1 residential deployments.
2. **Outdoor Unit (ODU) Commitment:**
    - Bidders must concurrently submit a firm, contractually committed product delivery roadmap for high-gain Outdoor Units (ODU, $\ge$ 12.0 dBi directional gain, IP67 weather-proofing, pole-mountable) within **6 to 9 months** following contract award.

4.2 CPE Warranty vs. Churned Reverse Logistics Lifecycle

1. **Manufacturer Warranty (Defect Liability Period):**
    - All supplied CPE hardware carries a mandatory **24-month (2-year) free replacement warranty** covering internal hardware defects, component burnout, and manufacturing faults.
2. **Churned CPE Recovery Rate ($\ge$ 70%):**
    - The contractual $\ge$ 70% recovery target applies exclusively to **reverse logistics operations executed by field force personnel** retrieving TelTech-owned CPE units from churned or canceled subscriber premises.
    - Recovered units undergo factory diagnostic testing, firmware re-flashing, cosmetic refurbishment, and re-packaging into active forward deployment inventory.
    - Hardware damage resulting from proven subscriber vandalism, fire, or force majeure is excluded from warranty and replenished via the Master Rate Card spare parts schedule.

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5. CIVIL, STRUCTURAL & SITE POWER INFRASTRUCTURE (LOTS 2 & 5)

5.1 Class T4 Greenfield Monopoles (103 Sites)

1. **Structural Wind Load Standards:**
    - All newly built Class T4 Greenfield Monopoles must be structurally engineered in compliance with the **National Structural Code of the Philippines (NSCP 2015, 7th Edition) / EIA/TIA-222-G**.
    - Structures must withstand extreme typhoon wind gusts of **270 km/h (Zone 2) up to 315 km/h (Zone 1)** under full design dead load:
    - 6x 5G Remote Radio Units (RRUs);
    - 1x Active Antenna Unit (AAU) or 3x directional passive panel antennas;
    - Up to **1.2 m² total projected windward surface area** for microwave backhaul antennas (equivalent to 1x 1.2m or 2x 0.6m high-wind parabolic dishes);

- Associated cable ladders, feeder runs, and top lightning protection rod.

2. **Foundation Hydrological Elevation:**
   - **Standard Non-Flood Soils:** Reinforced concrete equipment plinths shall have an elevation of **300 mm (0.30 m)** above finished grade level.
   - **Documented Flood Zones (50-Year Inundation History):** Civil contractors shall construct elevated structural steel equipment platforms raised **1.0 m to 1.5 m** above finished grade level, priced pursuant to the approved Master Rate Card.
3. **Starlink Satellite Backhaul Mounting Interface:**
   - Where hybrid LEO satellite backhaul is designated, contractors shall install a hot-dip galvanized mounting mast (minimum 60 mm / 2.375-inch outer diameter, 2.5 mm wall thickness) with an unobstructed view of the sky. For ground sites, a dedicated gantry standoff arm is utilized; for tower sites, a rigid leg standoff clamp is installed.

5.2 AC Electrical Utility Interconnection & Standby Power

1. **Service Entrance Capacity:**
   - AC service drops shall be sized for **100A, 230V single-phase (or 3-phase where commercially available)**.
   - The contractor's civil/electrical scope includes installation of the weatherhead, service mast, utility-grade meter base, manual transfer switch (MTS), and main AC distribution board with Class I+II Surge Protective Devices (SPD). Utility energization and power billing accounts are administered directly by TelTech.
2. **Site Power Rectifier Architecture (Lot 2):**
   - Each site requires one (1) high-efficiency outdoor-rated rectifier system with N+1 modular switchmode rectifier redundancy (e.g. 3x 50A modules providing 100A continuous N+1 DC output). Remote SNMPv3 / Modbus-TCP telemetry reporting into the centralized OSS is mandatory.
3. **Diesel Generator Fuel at Commissioning:**
   - Contractors are responsible for supplying diesel fuel strictly for generator commissioning and full-load burn-in testing (minimum 4 hours continuous run). Routine fuel replenishment transitions to TelTech O&M upon Provisional Acceptance (PAC).

5.3 Technical Site Survey Reports (TSSR) & Drone Sweeps

1. **Panoramic Photographic Standards:**
   - For tower co-location sites (Class T1–T3), contractors shall perform 360° panoramic photographic sweeps at antenna centerline height (and 20m, 30m, 40m AGL) using calibrated drone photographic platforms.
   - In areas where drone operations are legally prohibited by aviation authorities (CAAP restricted airspace, airport buffer zones, military reservations) or severe inclement weather, high-resolution manual panoramic photography captured by certified tower riggers from the tower climbing platform is accepted.

6. OPERATIONAL SLAS, RADIO NETWORK KPIS & SSV ACCEPTANCE (APPENDIX C)

6.1 Radio Performance Indicators (Nationwide 24/7/366)

To harmonize rigorous carrier-grade service quality with real-world FWA wireless propagation environments, the following SLA thresholds are formally enacted:

1. **Call / Session Setup Success Rate (CSSR):**
   - **Target Standard: 97.0%** (revised from 99.0%)
   - **Minimum Threshold: 96.0%** (revised from 97.5%)
   - Formula: $\text{CSSR} = \text{RRC Setup Success Rate} \times \text{5G QoS Flow Setup Success Rate} \times 100\%$
2. **Call Drop Rate (CDR):**
   - **Target Standard: $\le$ 1.5%** (revised from $\le$ 0.5%)
   - **Maximum Threshold: $\le$ 2.0%** (revised from $\le$ 1.0%)
3. **Packet Loss Rate:**
   - Evaluated at the **PDCP (Packet Data Convergence Protocol) Layer** pursuant to 3GPP TS 38.314.
   - **Target Standard: $\le$ 0.10%**; **Maximum Threshold: $\le$ 0.20%**.
4. **Time-Consistent Busy Hour (TCBH):**
   - Evaluated on a per-cell basis as the rolling continuous 60-minute window demonstrating highest combined uplink/downlink traffic volume over a 30-day observation period.

6.2 Service Credit Assessment & Stabilization Governance

1. **Financial Penalty Floor:**
   - Service Credit liquidated damages are assessed **exclusively when monthly performance breaches the Minimum Threshold floor** (or exceeds the Maximum Threshold for CDR/Packet Loss). Performance falling between the Target Standard and Minimum Threshold mandates engineering root-cause analysis and corrective cluster optimization without financial penalty.
2. **Post-PAC Cluster Stabilization Window:**
   - A **90-day stabilization grace period** is granted for each site rollout cluster commencing from Provisional Acceptance (PAC), during which financial Service Credits for Network Availability (NAV) and MTTR are suspended while operational baselines and field spares logistics are stabilized.
3. **OSS Throughput Counters vs. Contractual SLAs:**
   - Financial penalties shall not be assessed against instantaneous OSS cell-edge throughput counters due to dependencies on subscriber traffic distribution and concurrency. Cell-edge speed compliance is governed by Single Site Verification (SSV) and cluster drive acceptance.

6.3 Single Site Verification (SSV) Speed Criteria by RSRP Brackets

Single Site Verification stationary throughput benchmarks for a 60 MHz Band n50 channel under a standard 8:2 (DDDDDDDSUU) or 7:3 TDD frame configuration are established as follows:

- **Near Zone (RSRP $\ge$ -80 dBm):** Downlink $\ge$ 200 Mbit/s | Uplink $\ge$ 25 Mbit/s
- **Mid Zone (-95 dBm $\le$ RSRP < -80 dBm):** Downlink $\ge$ 100 Mbit/s | Uplink $\ge$ 10 Mbit/s
- **Cell Edge Zone (-105 dBm $\le$ RSRP < -95 dBm):** Downlink $\ge$ 35 Mbit/s | Uplink $\ge$ 5 Mbit/s

(Note: A 5 Mbit/s uplink floor is mandatory to guarantee TCP ACK throughput and interactive voice/video responsiveness).

6.4 Field Installation & Maintenance Metrics

1. **Subscriber Installation Scope:**
    - Encompasses on-site RF signal verification, optimal wall/roof/pole mounting, outdoor weather-sealed Cat6 cabling (up to 30m), PoE injection, indoor Wi-Fi router setup, zero-touch provisioning via FMS app, and verified speed test sign-off.
2. **First-Time-Right (FTR) Installation Rate ($\ge$ 95%):**
    - Calculated monthly as:

$$\text{FTR Rate} = \left(\frac{N_{\text{completed installations}} - N_{\text{repeat visits within 14 days}}}{N_{\text{completed installations}}} \right) \times 100\%$$

- Any physical revisit required within fourteen (14) calendar days following sign-off due to workmanship defects, misaligned antennas, or provisioning errors is classified as a failed FTR event.

1. **Spare Parts Management Service (SPMS) NBD Delivery:**
    - Next Business Day (NBD) replacement requires dispatch and physical delivery of replacement spare modules to the **designated regional forward maintenance depot / TelTech regional maintenance center** within the next business day (Monday through Saturday, cut-off 15:00) following RMA notification.

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7. DIGITAL OSS/BSS, FIELD FORCE AUTOMATION & VAS INTEGRATION (LOT 4)

1. **Native Automated Business Intelligence (BI):**
    - The BSS/OSS suite must incorporate an integrated reporting engine providing scheduled generation of operational, billing, subscriber churn, and SLA compliance reports in CSV, Excel, and PDF formats. Furthermore, open database views / secure REST APIs must be exposed for data pipeline ingestion into TelTech's enterprise data lake.
2. **Cloud Contact Center Integration (CCaaS):**

- The BSS/CRM platform must expose standard TM Forum Open APIs (**TMF621 Customer Management, TMF622 Product Ordering, TMF632 Party Management**) for single-pane CTI screen-pops, automated customer verification, and unified ticketing integration with TelTech's cloud contact center.

3. **Value-Added Services (VAS) & SMS Gateway Connectivity:**
   - The BSS must provide native **SMPP v3.4 and RESTful SMS gateway adapters** to trigger automated customer notifications (one-time passwords [OTP], billing invoices, payment acknowledgments, and service notifications). TelTech provides the SMS gateway termination service.
4. **Field Force Management (FMS) Mobile Application Capabilities:**
   - **Navigation:** Deep-linking or redirecting from within the FMS app to standard third-party navigation applications (Google Maps, Waze) for routing to site/subscriber coordinates is fully approved.
   - **Electronic Signatures:** Capturing touchscreen handwritten customer signatures on the mobile application upon job completion satisfies formal proof-of-delivery. Third-party digital certificate authorities (PKI/CA) are not required for residential installation orders.
   - **RF Signal Logging:** Certified mobile RF verification tools (e.g. Cellular-Z, G-NetTrack, Nemo Handy, or OEM diagnostic suites) are approved for logging RSRP, RSRQ, SINR, and Cell ID parameters into completion reports.
   - **Automated Service Provisioning:** Work orders successfully validated and signed off on the FMS app must automatically trigger real-time subscriber service activation in the BSS/OSS.
5. **Software SLA Boundary vs. Field Managed Services Scope:**
   - Standalone OSS/BSS software SLAs (Lot 4) govern application availability ($\ge$ 99.95%), transaction processing latency, and API throughput. Field execution metrics (Dispatch Lead Times, FTR Installation Rates, Churned CPE Recoveries) apply strictly to the Managed Services & Field Operations scope (Lot 5 / Appendix E) when awarded.

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8. COMMERCIAL PROPOSAL, 10-YEAR TCO & PROCUREMENT GOVERNANCE (DOCUMENT 06)

1. **Total Pricing Scope (5,000 Candidate Sites):**
   - Commercial proposals must provide firm, fixed unit rates and total pricing based on the complete **5,000 candidate site scope** (Lot 1 through Lot 7). Project deployment shall proceed under the multi-batch rollout roadmap (Phase 1: 1,000 sites; Batches 2–5: 1,000 sites each).
2. **Currency Conversion Benchmark:**
   - For bid evaluation and financial modeling, foreign currency (USD) and local currency (PHP) proposals shall be evaluated using the benchmark conversion rate of **USD 1.00 = PHP 55.00**. For offshore active equipment (Lots 1, 2, 3, 7), contracts and Letters of Credit (LC) may be denominated directly in USD.
3. **Value Added Tax (VAT) Status:**

- All unit rates, material pricing, and services fees in Document 06 must be quoted **exclusive of 12% Philippine Value Added Tax (VAT-Exclusive)**. VAT will be added as a separate billing item pursuant to Bureau of Internal Revenue (BIR) regulations.

4. **Partial Lot Bidding & Custom Line Items:**
   - Bidders are fully permitted to bid on single Lots or arbitrary combinations of Lots (e.g. RAN only, Core only, Civil only). Non-quoted lots shall be marked 'NO BID'. Custom proprietary hardware or services may be included utilizing the designated custom line-item entry rows.
5. **Years 6–10 OPEX in 10-Year TCO:**
   - Bidders may structure Years 6–10 extended maintenance and managed services pricing as either fixed annual fees or transparent percentage-based annual escalations within Form C of Document 06.
6. **Mandatory OEM Technical Training:**
   - Turnkey proposals must include comprehensive OEM technical training and certification programs covering 5G SA Core administration, Band n50 RF planning/optimization, and OSS/BSS operations for at least twenty (20) TelTech technical personnel.

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9. CONSOLIDATED RESPONSE MATRIX (PART 3: BIDDER CLARIFICATIONS Q045 – Q100)

No.	Domain Classification	Document Reference	Prospective Bidder Inc
1	**5G Core & IT Infrastructure (Lot 3)**	01_RFP_5G_FWA_Master_Tender_Document.pdf 1.2 ProgramLOT 3: CLOUD-NATIVE 5G CORE (5GC SA) & DATACENTER INFRASTRUCTURE	Please specify how muc million users in the first throughput will be need future.
2	**RAN Equipment & Access Hardware (Lot 1)**	07_RFP_5G_FWA_Candidate_Site_List_and_Rollout_Schedule.xlsx Phase 1 (First 1,000 Sites)	We have obtained the la column T in the table or specific latitude and lon provide these tower's la coordinates to support evaluation?

No.	Domain Classification	Document Reference	Prospective Bidder Inc
3	**RAN Equipment & Access Hardware (Lot 1)**	01_RFP_5G_FWA_Master_Tender_Document.pdf 2.1 Spectrum Physics: Band n50 Advantage	The current capability d range of 1512~1517 MH Bidder requires more de conduct an evaluation. I the NTC's demonstratio Contractor / Prospective information: the suppre signals in the 1432~145 suppression level requir 1517 MHz.

No.	Domain Classification	Document Reference	Prospective Bidder Inc
4	**Customer Premises Equipment (Lot 7)**	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 7: 5G CUSTOMER PREMISES EQUIPMENT (CPE - LOT 7)	There is no ODU N50 pr while for development. ' indoor N50 first

No.	Domain Classification	Document Reference	Prospective Bidder Inc
5	**Digital OSS/BSS & Automation (Lot 4)**	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Is there an existing BI/re confirm if BSS doesn't r anymore.
6	**Digital OSS/BSS & Automation (Lot 4)**	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Is there an existing cont separate customer ticke
7	**Digital OSS/BSS & Automation (Lot 4)**	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	How about VAS system? with existing SMS?
8	**Civil Works & Site Siting (Lot 5)**	01_RFP_5G_FWA_Master_Tender_Document LOT 5: RF PLANNING, SITE SURVEY & CLUSTER OPTIMIZATION	We note your confirmati is accepted. To align ex following: (a) TSSR photographic panoramic photography height from towerco. Dr TSSR. (b) The acceptance crit that they clearly depict environment (including potential obstructions, a sufficient for line-of-sig camera resolution, foca required; standard phot the site survey team is a

No.	Domain Classification	Document Reference	Prospective Bidder Inc
9	**Civil Works & Site Siting (Lot 5)**	05_Appendix_D_Site_Survey_RF_Planning_Civil_Power_Guidelines.pdf 4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD SITES)	For 103 sites, Do we ne foundation in the comm confirm the required he as VO.
10	**Civil Works & Site Siting (Lot 5)**	03_Appendix_B_Commercial_Proposal_BoQ_Templates TABLE 3: LOT 2 — POWER SYSTEMS & PASSIVE INFRASTRUCTURE	1. Please suggest the re the Contractor / Prospe handover the site to Tel 2. If Yes, When we hand need to refuel the fuel t
11	**Civil Works & Site Siting (Lot 5)**	05_Appendix_D_Site_Survey_RF_Planning_Civil_Power_Guidelines.pdf 2.4 Backhaul Transmission Path Survey	Do we need to provide a Starlink? If yes, please s for the pole.
12	**Civil Works & Site Siting (Lot 5)**	05_Appendix_D_Site_Survey_RF_Planning_Civil_Power_Guidelines.pdf 4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD SITES)	Please suggest how ma consider in the monopo provide the dimension a RRU、MW which need t
13	**Civil Works & Site Siting (Lot 5)**	05_Appendix_D_Site_Survey_RF_Planning_Civil_Power_Guidelines 03_Appendix_B_Commercial_Proposal_BoQ_Templates 2.3 Power Infrastructure & Utility Audit TABLE 3: LOT 2 — POWER SYSTEMS & PASSIVE INFRASTRUCTURE	For the New monopole s capacity we need to app

No.	Domain Classification	Document Reference	Prospective Bidder Inc
14	**Digital OSS/BSS & Automation (Lot 4)**	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Regarding the CPE reco Contractor / Prospective period allows for free re such a service acceptab cannot be controlled.
15	**SLAs, KPIs & Acceptance Standards (App C)**	04_Appendix_C_SLAs_KPIs_Liquidated_Damages.pdf 3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIS & SERVICE CREDITS	Regarding the MTTR、N Due to the lack of live n inputs, it is impossible t KPIs: SLA is recommended to actual values observed acceptance and commis
16	**Commercial Proposal & BoQ (Doc 06)**	06_RFP_Commercial_Price_Book_and_BoQ_Template All	The portal "3. Commerc contents is different wit "06_RFP_Commercial_F Kindly clarify which vers
17	**Commercial Proposal & BoQ (Doc 06)**	Portal "3. Commercial BoQ & Financing Lots 1-7" All	Regarding the vendor su "3. Commercial BoQ & F Since the current BOQ c project, please clarify th subsequent years: CAPEX: Assuming the P exact pricing be applica 5? OPEX: Since vendors ar Phase 1, how should we For Phase 2, do we mult Does this same logic ap to Year 5?

No.	Domain Classification	Document Reference	Prospective Bidder Inc
18	**Commercial Proposal & BoQ (Doc 06)**	Portal "3. Commercial BoQ & Financing Lots 1-7" All	As the USD to PHP exch significantly right now, I financial risk. We propo documents (BoQ, PO, a you prefer to pay in PHF exchange rate on the da
19	**Commercial Proposal & BoQ (Doc 06)**	Portal "3. Commercial BoQ & Financing Lots 1-7" All	Kindly clarify for Comm need to be filled online: 1. If we are unable to pr the bidder leave the uni 2. Where is the best pla assumptions?
20	**Service Level Indicators & Field Metrics**	01_RFP_5G_FWA_Master_Tender_Document LOT	What is installation? In v required? Is the Contrac responsible for the insta New Installation Dispat
21	**Service Level Indicators & Field Metrics**	01_RFP_5G_FWA_Master_Tender_Document LOT	How is the First-Time-R calculated?

No.	Domain Classification	Document Reference	Prospective Bidder Inc
22	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Since it is a managed se accept the OSS tools of How to handle the tools
23	SLAs, KPIs & Acceptance Standards (App C)	04_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/365 Nationwide)	Service creadit penalty threshold mean Minimu
24	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/366 Nationwide)	kindly provide formula , rate X erab assignment
25	SLAs, KPIs & Acceptance Standards (App C)	06_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/367 Nationwide)	busy hour means auto b such as 19:00- 20:00?
26	SLAs, KPIs & Acceptance Standards (App C)	07_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/368 Nationwide)	1. because there is " cel this drive test KPI ? use throughput? 2. how to define cell edg
27	SLAs, KPIs & Acceptance Standards (App C)	5.7 Radio Frequency (RF) Cluster Optimization	the target of cluster opt outdoor target , any targ throughput ?

No.	Domain Classification	Document Reference	Prospective Bidder Inc
28	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_D_Site_Survey_RF_Planning_Civil_Power_Guidelines.pdf 5.. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA	for Near , mid and far lo simulation of typcial site and give proper distanc RSRP level at -90, -100,
29	RAN Equipment & Access Hardware (Lot 1)	01_RFP_5G_FWA_Master_Tender_Document.pdf 2.1 Spectrum Physics: Band n50 Advantage	From our general under allocation are usually ur to know whether TelTec from the NTC to use 80 outside the NCR.
30	RAN Equipment & Access Hardware (Lot 1)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 1: RADIO ACCESS NETWORK (RAN) TECHNICAL SPECIFICATIONS	Antennas are generally positions on towers. Tov antenna's windward are TelTech have requireme area and wind load capa Prospective Bidder's ex base stations, it is recor area does not exceed 0.
31	CPE Warranty & Reverse Logistics	01_RFP_5G_FWA_Master_Tender_Document.pdf	Please confirm that the Period (DLP / Hardware equipment remains 24 r of PAC issuance per site clarify whether the Exte Management Service (S Years 6–10 for TCO calc separate, optional line it B Commercial BoQ.

No.	Domain Classification	Document Reference	Prospective Bidder Inc
32	OEM Technical Training & Enablement	01_RFP_5G_FWA_Master_Tender_Document.pdf	Please clarify whether f and certification progra n50 RAN, BSS/OSS, and TelTech's engineering a kindly provide the estim preferred training locati Manila/Clark vs. OEM he should be quoted as a c Commercial BoQ.
33	Commercial Proposal & BoQ (Doc 06)	Portal "3. Commercial BoQ & Financing Lots 1-7"	For the Years 6–10 Cum Services and Extended TCO evaluation, please to include an annual Co inflation indexation and account for long-term m
34	Commercial Proposal & BoQ (Doc 06)	Portal "3. Commercial BoQ & Financing Lots 1-7"	Are all unit prices in Cor VAT Ex (Value Added Ta
35	Power Systems & Facilities (Lot 2)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 2: POWER SYSTEMS & PASSIVE SITE INFRASTRUCTURE	1. Does it requires 2 Rec Configuration? For ex: 1 2. Do they need monito 3. Please confirm weath indoor or outdoor? 4. If outdoor installation same set up?
36	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	In the FM app, is it acce function redirects to an Maps?
37	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	In the FM app, is the e-s handwritten signatures? used for strong authent to be integrated. Currer

No.	Domain Classification	Document Reference	Prospective Bidder Inc
38	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	For RF signal verificatio professional software su accurate and professior
39	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Does the installation orc
40	RAN Equipment & Access Hardware (Lot 1)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 5: RF PLANNING, SITE SURVEY & CLUSTER OPTIMIZATION	The areas covered by th mostly suburban or rura capability to use 5-mete to using 2D maps entire
41	RAN Equipment & Access Hardware (Lot 1)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 1: RADIO ACCESS NETWORK (RAN) TECHNICAL SPECIFICATIONS	According to the curren maximum average traffi exceed 1 Gbit/s. Therefc support 2 x 10GE + 2 x 1 recommended that only reserved, that is, 2 x 25
42	RAN Equipment & Access Hardware (Lot 1)	07_RFP_5G_FWA_Candidate_Site_List_and_Rollout_Schedule.xlsx Phase 1 (First 1,000 Sites)	Considering there are 8 advisable to use cabine
43	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Regarding the CPE reco Contractor / Prospective period allows for free re such a service acceptab cannot be controlled.

No.	Domain Classification	Document Reference	Prospective Bidder Inc
44	SLAs, KPIs & Acceptance Standards (App C)	04_Appendix_C_SLAs_KPIs_Liquidated_Damages.pdf 3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIS & SERVICE CREDITS	Regarding the MTTR、N Due to the lack of live n inputs, it is impossible t KPIs: SLA is recommended to actual values observed acceptance and commis
45	Service Level Indicators & Field Metrics	01_RFP_5G_FWA_Master_Tender_Document LOT	What is installation? In v required? Is the Contrac responsible for the insta New Installation Dispatc
46	Service Level Indicators & Field Metrics	01_RFP_5G_FWA_Master_Tender_Document LOT	How is the First-Time-R calculated?
47	Service Level Indicators & Field Metrics	01_RFP_5G_FWA_Master_Tender_Document LOT	Please provide the proc recovery, and maintenar

No.	Domain Classification	Document Reference	Prospective Bidder Inc
48	SLAs, KPIs & Acceptance Standards (App C)	04_Appendix_C_SLAs_KPIs_Liquidated_Damages.pdf 3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIS & SERVICE CREDITS	Does NBD in SPMS mea Prospective Bidder's re
49	Digital OSS/BSS & Automation (Lot 4)	02_Appendix_A_Technical_Specifications_Compliance_Matrix.xlsx SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION	Can the standalone OSS decoupled from the foll Installation Dispatch Le Installation Rate, and Ch Rate.
50	Civil Works & Site Siting (Lot 5)	APPENDIX E: SERVICES STATEMENT OF WORK & MASTER RATE CARD 4. MASTER SERVICES PRICING SCHEDULE & COMMERCIAL RATE CARD	1. Please suggest on wh tower retrofitting works, to towerco to do retrofit 2. Tower retrofitting nee first, and this scope are case by case, Please su
51	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/366 Nationwide)	Suggest CSSR target 97 based on huawei global setup success rate 99% The CSSR further impro terminal compatibility is coverage, and transmis
52	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/366 Nationwide)	Suggest target 1.5%, r The same reason that K constrainted from termi extent of network cover indoor CPE.
53	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/366 Nationwide)	suggest to remove both user behavior and user already reflected in SSV

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54	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/366 Nationwide)	There is no OSS KPI late are available, which hav
55	SLAs, KPIs & Acceptance Standards (App C)	05_Appendix_C_SLAs_KPIs_Liquidated_Damages 3.1 Core Radio KPIs (Calculated 24/7/366 Nationwide)	suggest target 0.1%, 0.
56	RAN Equipment & Access Hardware (Lot 1)	05_Appendix_D_Site_Survey_RF_Planning_Civil_Power_Guidelines.pdf 5.. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA	from last time clarificati SSV SPEED TARGETS P - Near Zone (< 500m): I Mbps. - Mid Zone (1.0 – 1.5 km Mbps. - Far / Cell Edge (>= 2.0 Mbps (minimum SLA thr Suggest to be : Slot configuration: DL:U static test, >-80dBm: DL 200Mbps [-80~-95)dBm: DL 100M [-95~-105)dBm: DL 35M
57	Backhaul & Transmission Interconnect	01_RFP_5G_FWA_Master_Tender_Document 2.3 Transmission & Backhaul Topology	1, The SFP of Transmiss with BBU will be provide Prospective Bidder or T 2, The SFP of BBU will b Module, the SFP of tran also, or specific ones ar confirm the transmissio 3, The related configura for BBU integration will Contractor / Prospective

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End of Bid Bulletin No. 2.1. Promulgated as an Official and Binding Addendum by the Special Bids and Awards Committee (SBAC) / Procurement Directorate.